

Master Thesis Preliminary Results

Asymmetric-granularity friction in Spanish electricity markets

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Recap and IO question

Recap: Spain's 2024–2025 reform sequence

- Three reform stages staggered the move from MTU60 (hourly) to MTU15 (quarter-hourly):
 - Dec 2024:** ISP15 (imbalance settlement)
 - Mar 2025:** ID15 (intraday trading)
 - Oct 2025:** DA15 (day-ahead trading)
- This created a **10-month asymmetric-granularity window** (Dec 2024 – Sep 2025) where settlement was 15-min while DA cleared at 60-min.
- Plus the **April 28 2025 blackout** during DA60/ID15 – a confound we have to account for.

Period	ISP	DA	ID
Pre-Dec 2024	60	60	60
Dec24–Mar25 (ISP15-win)	15	60	60
Mar25–Sep25 (DA60/ID15)	15	60	15
Oct25 onward (DA15/ID15)	15	15	15



From the February proposal to today's findings

- In February I proposed a **theoretical extension of Ito and Reguant (2016)**, predicting:
 - **DA15** is the key reform that smooths imbalances.
 - **Dispersion** risk concentrates in the transitional DA60/ID15 window.
 - Finer granularity creates winners and losers across heterogeneous portfolios.
- Today: all three predictions are **empirically confirmed** – but the surviving microfoundation is system-layer settlement-clock mismatch, **not** the firm-level Allaz–Vila commitment-value channel.

Today's IO question

When the regulator changes settlement and trading clocks asymmetrically, **who pays the resulting friction, who captures the rent, and is the rule incentive-compatible** with marginal social cost?

Data and empirical strategy

Data and empirical strategy

- **OMIE bid-level data (Detalle de las ofertas)** – 6 yrs, 3-month publication lag (current to Jan 2026).
- **ENTSO-E TP**: A85 imbalance prices/volumes; A87 financial balance; A73 per-unit dispatch; A75 wind+solar; A65 D-1 load; A11 cross-border flows.
- **ESIOS (REE)**: liquicomun settlement bundle (181 families, 4.4M rows post-extension); per-BSP aFRR settlement; restrictions/redispatch.
- **Controls**: AEMET weather, CO₂/fuels, reservoir filling, installed capacity, French DA prices for the cross-country placebo.

Empirical strategy (per Feb deck):

- No clean external counterfactual – most European markets adopted MTU15.
- **Same-calendar-month pre-IDA baseline** + bootstrap CIs.
- **Cross-country placebo** (Spain vs France).
- Blackout-split robustness (PRE/POST 2025-04-28).
- OVB-cleaning under good/bad-control distinction.

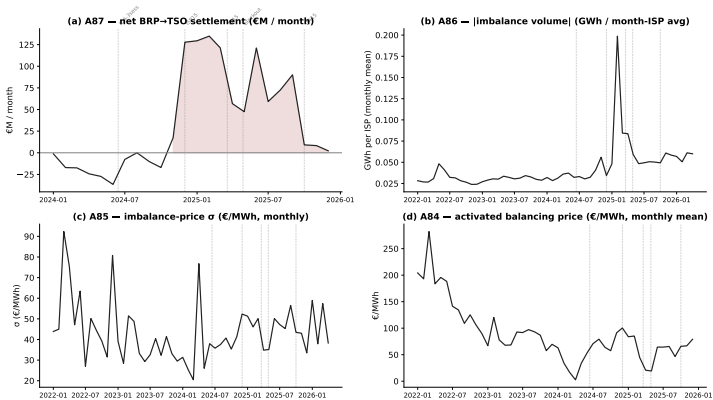
Status

The 3-month OMIE bid-data lag limits firm-level analysis post Jan 2026; today's system-layer findings

Findings: system-layer reform impact

Result 1 – Four ENTSO-E system metrics jump concordantly at ISP15

Four ENTSO-E system metrics jump concordantly at ISP15, moderate at MTU15-DA (S5)



- Monthly system metrics with calendar-month FE, regime contrasts vs same-calendar pre-IDA baseline (78 months).
- **A87** net imbalance settlement income.
- **A86** mean $|V_{imb}|$.
- **A85** imbalance-price σ .
- **A84** mean activation price.

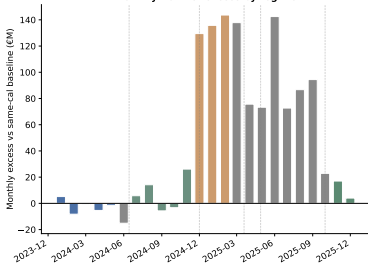
Joint null

The four metrics jump concordantly at ISP15 and moderate at MTU15-DA – this isn't one outlier metric.

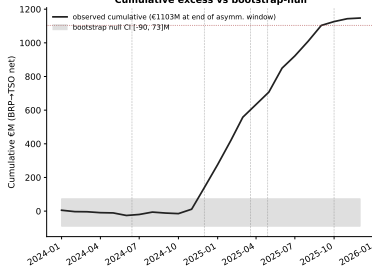
Result 2 – The headline: **€1,094.9M BRP→TSO settlement transfer**

S6 — Asymmetric-granularity window: cumulative BRP→TSO transfer $\approx$ €1.1B ($\approx 15\times$ upper bootstrap bound)

Monthly A87 net excess by regime



Cumulative excess vs bootstrap-null



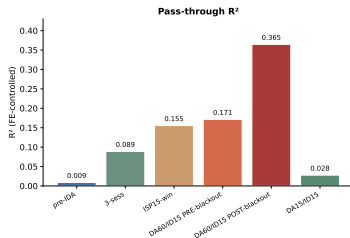
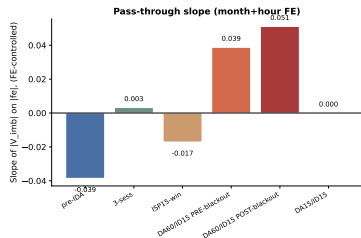
- **€1,094.9M** cumulative excess (A87 net) over the 10-month asymmetric window vs same-cal pre-IDA baseline.
- Bootstrap null CI $[-90, +73]$ **M€**; observed $\approx 15\times$ upper bound.
- Excess collapses to **€7.4M/mo** (8% of DA60/ID15 level) at MTU15-DA.

Caveat

This is a settlement **redistribution** (BRP→TSO, recycled to consumers via tariff with ~ 1 -yr lag), **not a deadweight loss**. Welfare interpretation requires a counterfactual on tariff pass-through (out of scope today).

Result 3 – Mechanism: forecast-error → imbalance pass-through

B6 — Forecast-error → imbalance pass-through: peaks DA60/ID15 POST-blackout, collapses at MTU15-DA



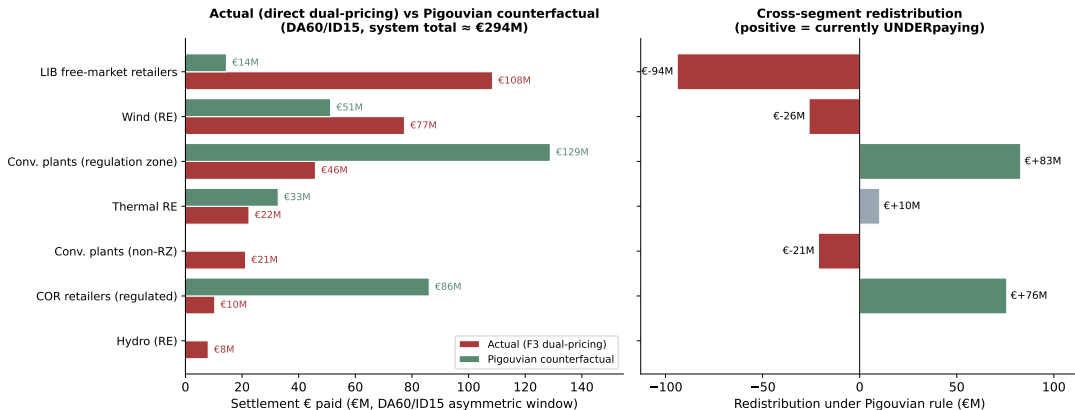
- Slope of $|V_{imb}|$ on $|wind+solar\ forecast\ error|$, month + hour FE.
- **Pre-IDA-late:** $R^2 = 0.023$ (clean baseline).
- **DA60/ID15 PRE-blackout:** $R^2 = 0.171$ (**7**× clean reform).
- **DA60/ID15 POST-blackout:** $R^2 = 0.365$ (16×, blackout amplification).
- **DA15/ID15:** $R^2 = 0.028$ – **the cleanest reform signature.**

Reading

Under asymmetric clocks, BRPs cannot match within-hour forecast errors → they propagate mechanically into imbalance volumes

Result 4 – Pigouvian incidence: who actually pays?

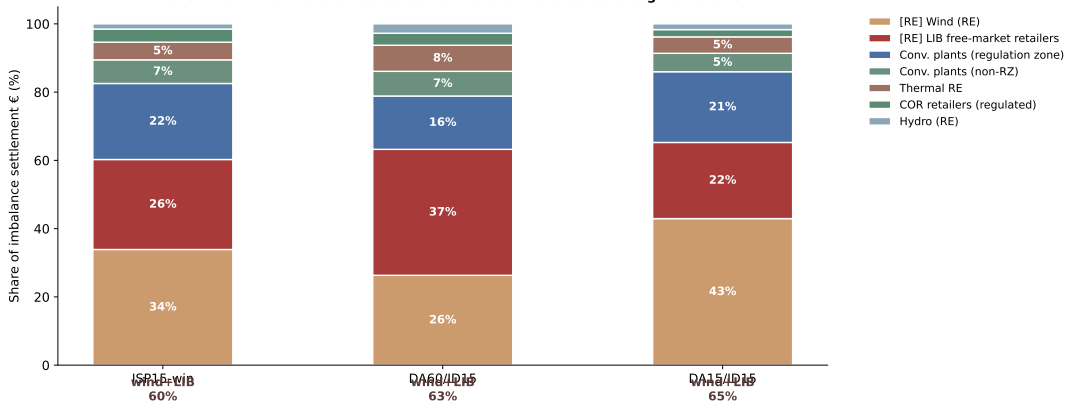
S7 — Pigouvian incidence in the asymmetric window (direct F3 dual-pricing decomposition)
LIB retailers + wind overpay ~€141M | Conv-RZ + COR + hydro RE underpay ~€159M | rule structurally favours dispatchable segments



- Direct dual-pricing decomposition: per-segment € from signed imbalance $\times \frac{prdvbaqh}{prdv suqh}$; reproduces 78% of system total ($\rho = 0.93$).
- **Pigouvian counterfactual:** charge each segment its β marginal contribution to system imbalance cost.

Result 5 – The structural finding: regime-invariance

Wind + LIB free-market retailers consistently pay 60-65% of imbalance settlement € across every post-ISP15 regime — MTU15-DA shrinks the total but does NOT relieve the renewable-segment share

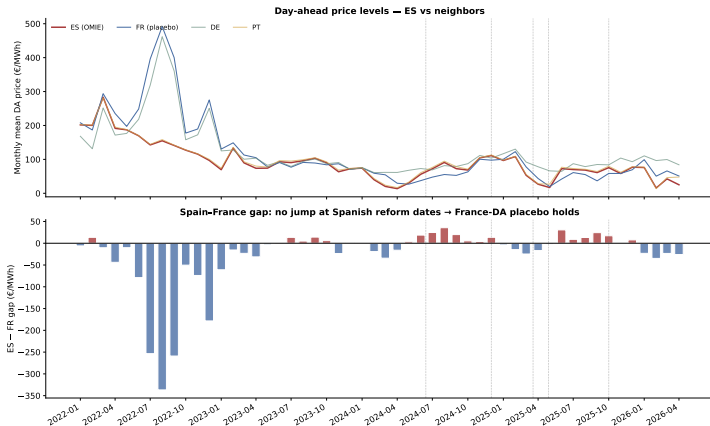


The IO bite

Wind + LIB free-market retailers consistently pay 60–65% of imbalance settlement € in EVERY post-ISP15 regime, including post-MTU15-DA. Clock-symmetry shrinks the SCALE of the redistribution but does not relieve its STRUCTURE.

Result 6 – Cross-country placebo (the Feb proposal said wasn't yet available)

B7 — France DA placebo: FR prices show no Spanish-reform-date jump (cross-country control)



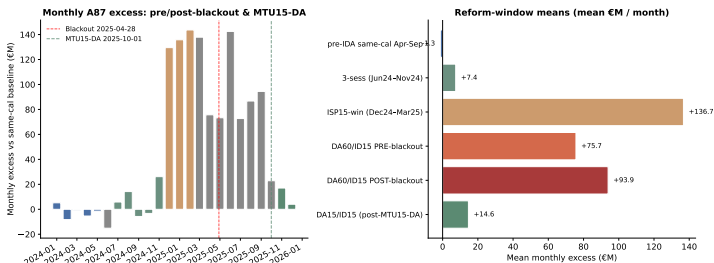
- Within-day DA price standard deviation by regime, Spain vs France.
- **Spain responds 2–3× more than France** across reform dates ($\Delta_{ES} = +13.7$ to $+19.7$ €/MWh; $\Delta_{FR} = +5.6$ to $+8.8$).
- France didn't undergo MTU15 in our window $\Rightarrow$ Spain-specific shock identified.

Caveat

The post-2025-04-28 blackout was also Spain-specific. Blackout-split (next slide) addresses this.

Robustness – the asymmetric-friction story is reform-driven, not blackout-driven

S6 robustness — DA15/ID15 collapse persists DESPITE operación reforzada (blackout is amplifier, not source)



- **Clean April 2025** (PRE-blackout, single month): **+€75.7M** excess.
- **May–Sep 2025** (POST-blackout DA60/ID15): €93.5M/mo (+24% vs clean April – modest amplification, not source).
- **Oct–Dec 2025** (DA15/ID15, post-blackout still in effect): **€7.4M/mo**.

Key signature

The DA15/ID15 collapse holds **despite** operación reforzada in effect – friction is reform-driven, not blackout-driven.

Synthesis

Two policy levers, two distinct mechanism-design failures

Lever 1 – Clock-symmetry

Failure: asymmetric clock scales create a BRP→TSO transfer of **€1.1B** over 10 months.

Lever: clock-symmetry at MTU15-DA.

Effect: reduces total scale $6\times$ (€91M/mo → €7.4M/mo).

Identification: same-calendar-month pre-IDA baseline + bootstrap (Result 2); microfoundation from forecast-error → imbalance volume pass-through R^2 collapse (Result 3); cross-country placebo (Result 6); blackout robustness (Result 7).

Lever 2 – Settlement-rule redesign

Failure: non-Pigouvian uniform-rate allocation across heterogeneous-MC segments.

Lever: settlement-rule redesign (Pigouvian per-segment pricing).

Status: not addressed by clock-symmetry. Wind + LIB retailers pay 60–65% of imbalance € in every post-ISP15 regime, including post-MTU15-DA (Result 5).

Implementation challenges: real-time per-segment marginal-cost measurement; trade-off with BRP forecast-accuracy incentives. Open mechanism-design problem, not a clean policy intervention.

What this talk does *not* show

- Today's findings are the **system-layer reform-impact slice**. The thesis maps three additional IO channels covered in Parts II–IV:
 - **Firm-level structural market power** – IB DA cleared-price-difference rent $\sim$ €820M post-MTU15-IDA, regime-invariant. Hydro Cournot-pivotality at 63% of post-MTU15-IDA hours ([Bushnell, 2003](#); [Hortaçsu and Puller, 2008](#); [Ciarreta and Espinosa, 2010](#)).
 - **Cross-market firm specialisation** – IB→DA hydro Cournot; **GE→aFRR** (€13.8M revenue post-MTU15-DA, +52% above IB); **Naturgy→CCGT generation** post-blackout (+7.1pp share gain).
 - **Post-CNMC strategic-availability conduct** – 13-yr CNMC enforcement record; replication of the 2023 SBO3 three-situation pivotality test shows Naturgy fleet-wide bid-price wedge of 11–35% in zone-pivotal hours; SBO3 itself still +14% post-sanction ([Joskow and Kahn, 2002](#)).
- These findings **pre-date and survive** the MTU15 reform – they don't fit the reform-impact frame of this talk.

Conclusions and next steps

Conclusions

Spain's MTU15 reform sequence is **empirically tractable** via same-cal pre-IDA baseline + cross-country placebo; the Feb-deck [Ito and Reguant \(2016\)](#) predictions hold at **€1.1B order of magnitude**.

The reform reveals **two distinct mechanism-design failures**:

- **Lever 1 (clock-symmetry)**: implemented at MTU15-DA, reduces scale 6×.
- **Lever 2 (settlement-rule redesign)**: open, structural correlation between renewable forecast errors and system imbalance direction (**LIB retailer correlation 0.74–0.81**) means renewables systematically face the penalty side of dual pricing.

The asymmetric-granularity friction **is reform-driven, not blackout-driven**: clean PRE-blackout April 2025 alone is +€75.7M; post-MTU15-DA collapse holds despite operación reforzada.

Next steps

- **Structural model extension:** distinguish *scale* (clock-symmetry) from *structure* (Pigouvian incidence) levers; quantify implementation trade-offs of segment-conditional pricing.
- **Brodersen-style Bayesian structural time series counterfactual** ([Brodersen et al., 2015](#); [Märkle-Huß et al., 2018](#)) as a complementary identification strategy for the next draft.
- **Per-BRP imbalance settlement verification** – ESIOs server has been returning HTTP 500 for 2 weeks on the per-BRP cost archive; will retry. Reduced-form per-segment evidence (today's Results 4–5) is interpretable as upper-bound on cross-segment redistribution.
- **Thesis chapters Parts II–IV** – firm-level structural market power, cross-market specialisation, post-CNMC strategic-availability conduct (separate from today's reform-impact frame).
- **Master thesis structure proposal** drafted; 5-part synthesis of 37 alive findings.

Pipeline status

Data pipeline current to April 2026: 4.4M-row `liquicomun_all` (ESIOS, 181 families); 5M individual mFRR bid records (ENTSO-E, just unlocked via TP File Library); cross-country French imbalance prices integrated.

Thank you!

Questions and discussion welcome.

References

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